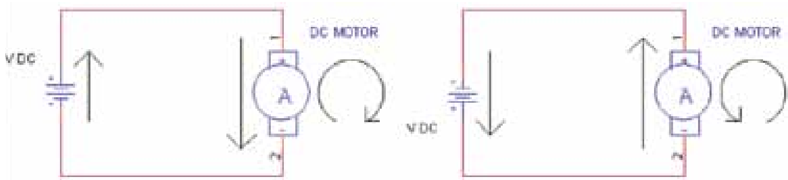
internal wiring of some DC motors. These factors may be present in the form of manufacturing defects, and spotting them by just looking at the motor is very difficult. The best way to check it is to power the motor with a suitable battery. Apply the power leads from the motor to the terminals of the battery or supply and note the direction of the rotation of the shaft. When you reverse the power leads from the motor, the motor shaft should rotate in reverse direction.

DC motors can be either continuous or stepper. In a continuous DC motor, the application of continuous power causes the shaft to rotate continually, while in a stepper the application of power causes the shaft to rotate a few



degrees, then pause, and then rotate again. In the continuous motor, the shaft stops only when the power is removed or if the motor is stalled, which means that the load is too much for the motor and it cannot drive it. Each 'step' of rotation of a stepper motor is precisely of a fixed angle of rotation.

Motor specifications

There are four parameters, which you should consider while buying a motor for yourself - voltage, current draw, speed, and torque.

Operating (or rated) voltage

Operating voltage gives the value of voltage which when applied across its terminals, produces best efficiency with a small DC motor, the rating is usually a range, like 1.5 to 6 V. Some high-quality DC motors are designed for a specific voltage, such as 12 or 18 V. But the kinds of motors of most interest to robot builders are the low-voltage variety—those that operate at 1.5 to 12 V.

Most motors can be operated satisfactorily at voltages higher or lower than those specified. A 12 V motor is likely to run at 8 V, but it may not be as powerful as it could be and it will run slower. If you use very less voltage (around under 50 percent of the specified rating) as compared to its rated voltage, the motor will stop working. Also, a 12V motor is likely to be able to run at 16 V. As the voltage across its terminal increases, speed of the shaft